

Collaborative VR: Conveying a Complex Disease and Its Treatment

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Figure 1: Excerpts of the virtual environments the participants explored with experts. The participants' names were removed.

ABSTRACT

In medical research, there are constantly new findings that lead to new insights and novel treatments. Hence, physicians always need to stay up-to-date to provide high-quality patient care. Virtual reality is unique among the available learning methods as it offers the opportunity to collaboratively explore the inside of the human body to experience and understand the complex workings of medical interventions. We realized such a VR-Learning-System and assessed its potential with medical experts ($n = 9$). Initial results indicate that experts prefer the new system, and according to their subjective assessment, they achieved a higher learning outcome compared to conventional methods.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Collaborative interaction; Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality

1 INTRODUCTION

It is crucial for medical professionals to stay up-to-date on rapid advances in their field to provide good medical care to their patients. This usually includes countless hours reading papers, clinical reports, and medical journals. But this modality is not a good fit for understanding the complex processes taking place inside the human body. Here, virtual reality (VR) offers the possibility to interactively convey complex subjects in a simple and detailed way and experience them up close. Therefore, in this contribution, we explore the potential of an alternative method for knowledge transfer with interactive VR. Its novelty is that several experts from medicine and pharmacy interactively jointly walk through a patient case experiencing and discussing all details, from the symptoms to the treatment, regardless of location. The following contributions have already highlighted the relevance and potential of different application areas

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of VR in medicine. This technology can be used, e.g., to transfer knowledge in a standardized and rapid way [2]. Initial results also indicate that the complex priming procedure of a medical device with VR can achieve a higher learning outcome than conventional methods [1]. Furthermore, telepresence allows a remote expert to collaborate in VR with a local user through augmented reality during a medical intervention [5]. Findings also reveal that VR can improve paramedic training where two trainees collaboratively treat a patient with anaphylactic shock [4]. At the same time, an instructor interactively controls the health status of the virtual patient.

2 SYSTEM DESIGN

This system allows several people to interact and communicate with each other simultaneously from different physical locations. They explore different virtual scenarios in which the complex and rare disease Atypical Hemolytic Uremic Syndrome (aHUS) is reproduced, explained, and discussed based on a real patient case. The storyline is structured so that a patient first describes his symptoms. Then a doctor explains the course of the disease and the associated problems during diagnosis. After the disease has been diagnosed, the doctor describes the disease to the patient, who consequently reveals his associated psychological and social challenges. Afterward, a pharmacy expert explains the mode of action of the necessary pharmaceutical therapy. These people representing the doctor and the pharmacy specialist are real experts who practice this profession in real life. They are able to answer and discuss the questions of the participants during the entire scenario in a professional manner. Apart from that, a special feature of this method is that the presenters and participants interact in different virtual environments (see Fig. 1) that would be impossible in reality. Initially, they virtually meet in front of a hospital, then walk through the different wards of the hospital, discuss the disease in the doctor's office, investigate the disease's mode of action and treatment in the patient's bloodstream as well as the corresponding triggers of the disease in the genetic material. This application is designed and implemented in cooperation with the companies Alexion Pharma, Onlive, and Arthur. The collaborative software is implemented with the Unity engine and the Photon networking framework. As shown in Fig. 1, all participants, including their movements, are represented as avatars.

Table 1: Questions participants rated after participation using a 5-point Likert scale ranging from 1 = “do not agree at all”, to 5 = “fully agree”.

ID	Question
Q1	The handling of the technical equipment went without problems.
Q2	Discomfort symptoms such as fatigue, nausea, dizziness, or headaches did not occur.
Q3	Overall, the VR experience was very pleasant.
Q4	The communication and sense of togetherness with the other participants was improved compared to a meeting, e.g., via Zoom or Microsoft Teams.
Q5	Compared to conventional methods (e.g., literature, presentation), this approach offers a higher learning outcome.
Q6	Compared to conventional methods (e.g., literature, presentation), I prefer this way of imparting knowledge.
Q7	There is interest in such an event in our clinic.
Q8	This VR scenario would help to convey complex diseases such as aHUS more effectively to patients.

To not overwhelm the participants with interactions, they can only move by using the teleportation technique. Moreover, they can also mute their microphone. The presenters also have the option to move objects and trigger animations.

3 USER STUDY

In order to investigate how medical experts assess this method in terms of experience, training success, and its associated potential, we conducted an experiment. For this purpose, nine medical experts were recruited from different medical institutions. All these experts are nephrologists since this field is responsible for treating aHUS. Their age ranged between 22 to 50 ($M = 36.44$, $SD = 10.22$), and they had a mean self-reported expertise with VR of 2.9 (from 1 = “no experience”, to 5 = “much experience”). Concerning the procedure, we first instructed the participants in handling the hardware (Oculus Quest 2) and software. Subsequently, each of them was placed in a separate cabin with a size of about 3×3 meters, which was only equipped with a chair. This separation was to prevent the participants from distracting or injuring each other [3]. There, they completed the VR scenario and subsequently assessed it by rating the questions listed in Table 1 using a 5-point Likert scale. In total, the participation took about 75 minutes and each participant agreed to the consent form.

4 RESULTS

We investigated the participants’ ratings with the non-parametric one-sample Wilcoxon test ($\alpha = 0.05$) since the ratings were not normally distributed and the sample size was small. The test revealed that the results of all eight questions were significantly above the neutral value of 3. The summarized results of the questionnaire from Table 1 are depicted in Fig. 2, indicating that the VR scenario performs well. Furthermore, the question “Which impressions do you remember most?” was only answered by five of the nine subjects; unanimously, it was the bloodstream depicted in Fig. 1.

5 DISCUSSION

Questions 1 to 3 indicate that the participants had no issues handling the technical equipment or cybersickness, and the overall VR experience was perceived to be very pleasant. Questions 4 to 8 demonstrate the potential of this technology compared to conventional methods. VR increased the communication and feeling of togetherness with the other participants as well as the learning success. Therefore, the subjects prefer this way of imparting knowledge and are interested in such an event in their clinic. Apart from that, almost every participant is convinced that this scenario would help convey complex diseases such as aHUS more effectively to patients. It also revealed

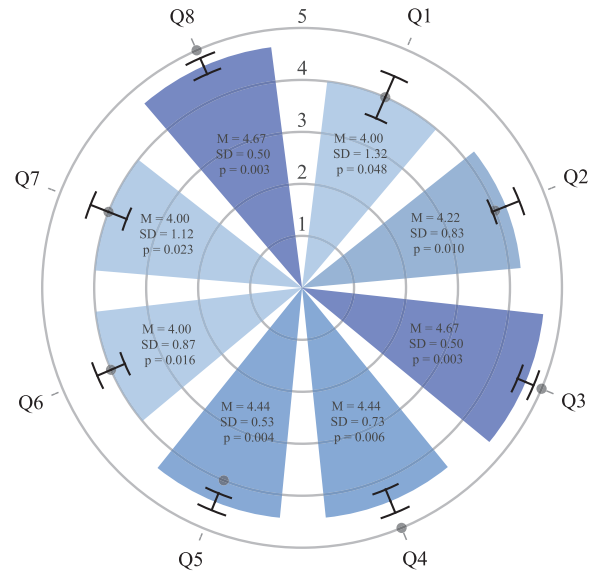


Figure 2: Results of the average subjectively rated questions from Table 1. Error bars depict the standard error, and the dots the median. Higher ratings represent a better result.

that the participants remembered the abstract impressions in the virtual bloodstream best, indicating that abstract spaces could increase interest without distracting attention.

6 LIMITATIONS

The study considered the potential of VR based on subjective ratings. Therefore, this VR scenario should be compared with a conventional method by using a subsequent test to obtain more meaningful results regarding the learning outcome. Due to the small number of participants, the results should only be treated as a tendency rather than solid evidence.

7 CONCLUSION

In this paper, we presented a collaborative VR system that potentially allows a variety of medical audiences to understand a complex and rare disease and its therapy. A unique advantage of this method is that the various experts’ discussions highlight the pharmaceutical treatment’s positive and negative aspects more clearly than possible in a one-on-one conversation with a consultant. Consequently, this method can improve patient care from diagnosis to therapy.

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